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How can slip stick friction be maximized between finger and computer mice

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August 11, 2026

Contents

1	Introduction	1
2	Background	2
2.1	Friction Phenomena	2
2.1.1	Viscous Friction	2
2.1.2	Static Friction	2
2.1.3	The Stribeck Effect	2
2.1.4	Rising Static friction	3
2.2	State Space Equations	3
2.3	The Dahl Model	3
2.4	Lugre Model	5
3	Experiment	5
3.1	Drag clicking specific methods	5
3.2	Research Hypothesis and Variables	6
3.3	Method	6
4	Results and Discussion	7
4.1	Observations	7
4.2	Limitations of the data	8

1 Introduction



Friction is a topic that has intrigued people for millenia, and pioneered to this very day. Think honestly, what causes friction? Small microscopic asperities? How does perfectly smooth rubber still present friction? Or jagged ice is so slippery? Is water the issue? Then why can you tread on water? From scientists, control engineers, and frankly, scholars that experienced its' effects in other forms, all trying to understand this confounding concept. Namely, slip-stick friction is the **limit cycle of an object**, continuous oscillation between slipping and sticking, such as seen in violins, door's creaking, badly engineered anything. It is a topic fixated by engineers in trying to reduce, common solutions include adding grease, oil, etc. However, in the realm of competitive Minecraft, there exists many instances where the desirable trait of clicking fast can be accompanied by exploitation of slip stick atop the mouse. This is referred to in the community as "drag clicking".



In this paper, we will attempt to find ways to test the slip stick properties pertaining to drag clicking by modeling material properties of common methods using advanced friction models such as LuGre and Dahl friction models.



2 Background

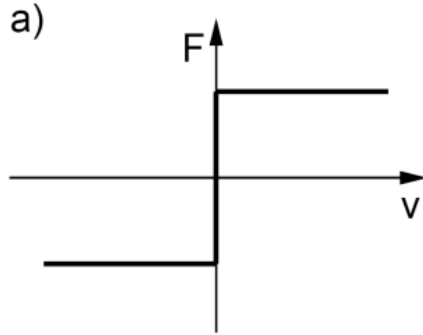


Figure 1: The change of friction with respect to velocity



Our commonly conceived model of friction—the coulomb friction model—has several limitations. It suggests that upon motion the frictional force is linearly proportional to the normal force and of the opposite direction of the velocity [4]. illustrated in figure 1. This is perceived as due to microscopic asperities interlocking with one another. This, for instance explains the independence of apparent surface area and dependence on normal force (i.e. the normal force pushes more asperities in contact, despite small appearant contact area, true microscopic contact area is much larger due to the increased force) However, friction still has much greater depth covering phenomena such as:

1. The Stribeck Effect
2. The Microscale pre-slip deformations
3. Dependence on past events
4. Ultimate Friction aging

2.1 Friction Phenomena

2.1.1 Viscous Friction

Viscous friction introduces the role of velocity in that sliding friction increases with velocity. Though exact instantaneous velocity cannot be provided via discrete data points, a treadmill setup could mitigate this with a constant velocity upon contact



2.1.2 Static Friction



Friction exhibits a noticable decrease when set in motion. Increasing applied force beyond this is usually the exciter for slip-stick friction.

2.1.3 The Stribeck Effect

The Stribeck Effect suggests that static friction does not suddenly jump to viscous friction but happens gradually. The rate of friction mode change depends much on material properties and in particular, we expect sharper jumps to exhibit superior drag-clicking capabilities



2.1.4 Rising Static friction

Refers to the increase in static friction as a result of time. Described as the normal forces gradually increasing true surface area between microscopic asperities over time.

2.2 State Space Equations

To coherently account for behaviour involving state changes, we must shift to dynamic models that uses state space equations. Most notably, the Dahl Model and LuGre Model. In state space equations, there exists "variables" that capture internal states of the system, when there is more than one, it is packaged into a vector matrix. State space equations define the derivative of state variable as a function of state variables, this gives us the rate of change at an instant, as we integrate it over small discrete units of time it gives us a prediction of the current state (current position, force, etc.). The predicted state variables are fed back into the derivative as linear equation constants to be then integrated again.[5] However, looking at the relationship between variables in a state space equation can describe the stability of a system, which is particularly useful in stick-slip motion.

2.3 The Dahl Model

While the standard textbook definition for friction stem from rigid body microscopic asperities, the Dahl Model instead visualizes two bodies making contact from elastic bristles scattered over the contacting surface[2]. Illustrated in figure 2.

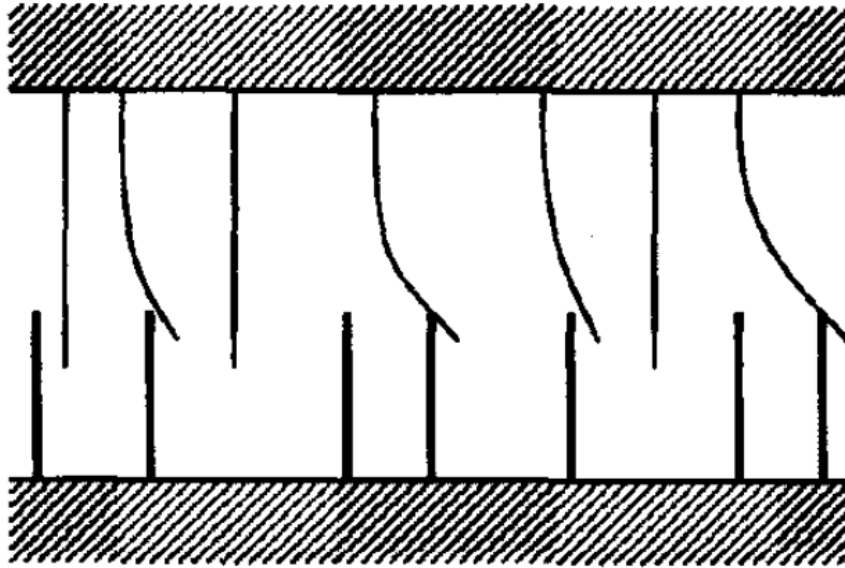


Figure 2: Depiction of Dahl Model's Bristles [2]

The Dahl model is defined as the following:

$$F = F_c \left(1 - e^{-\frac{\sigma_0 |x|}{F_c}} \right) \operatorname{sgn} \left(\frac{dx}{dt} \right) \quad (1)$$

Where F_c is the coulomb friction of the surface, x is the displacement from original position (this is primarily focused on the microscopic displacements), and σ_0 is the "stiffness" of microscopic bumps and affects the shift to maximum coulomb friction.

To use this for state space equations we must find the derivative using chain rule:

$$\frac{dF}{dx} = \cancel{F_c} \cdot e^{-\frac{\sigma_0|x|}{F_c}} \cdot \cancel{\frac{\sigma_0}{F_c}} \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (2)$$

$$= \sigma_0 e^{-\frac{\sigma_0|x|}{F_c}} \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (3)$$

and to put the actual current force state back into this derivative function (since state space equations depend on previous states) we rearrange eq (1) to isolate $e^{-\frac{\sigma_0|x|}{F_c}}$ like such:

$$F = F_c \left(1 - e^{-\frac{\sigma_0|x|}{F_c}}\right) \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (4)$$

$$\frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) = 1 - e^{-\frac{\sigma_0|x|}{F_c}} \quad (5)$$

$$e^{-\frac{\sigma_0|x|}{F_c}} = 1 - \frac{F}{F_c} \operatorname{sgn}\left(\frac{dx}{dt}\right) \quad (6)$$

and putting it back in the differential equation we get:

$$\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c}\right) \operatorname{sgn}\left(\frac{dx}{dt}\right) \operatorname{sgn}(x) \quad (7)$$

$$= \sigma_0 \left(1 - \frac{F}{F_c}\right) \operatorname{sgn}(x) \quad (8)$$

The Dahl model's equation shares similar properties of an exponential function take of the stress-strain curve (depicted in figure 3), most notably, the point at which it shifts from an elastic spring to continual bristle movement

Most importantly the Dahl Model introduces $z = \frac{F}{\sigma_0}$ as a state variable that represents the "average bristle deflection" [46]. This is very similar to Hooke's law. rearranging, we get:

$$F = \sigma_0 z \quad (9)$$

similarly, in Hooke's law presented in the Physics formula booklet:

$$F = kx \quad (10)$$

Differentiating $z = \frac{1}{\sigma_0} \cdot F$ to find the state equation:

$$\frac{dz}{dt} = \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot \frac{dx}{dt} \quad (11)$$

$$= \frac{1}{\sigma_0} \cdot \frac{dF}{dx} \cdot v \quad (12)$$

$$(13)$$

and with $\frac{dF}{dx} = \sigma_0 \left(1 - \frac{F}{F_c} \operatorname{sgn}(v)\right)$

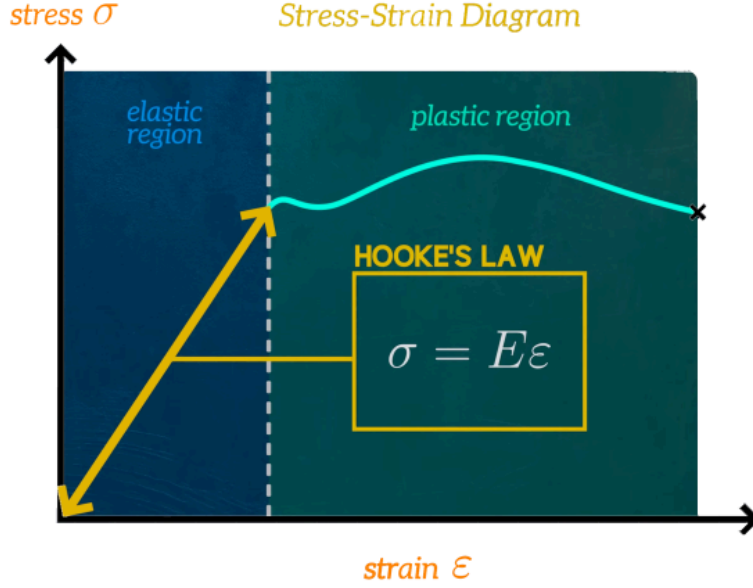


Figure 3: An annotated stress-strain curve[3]

$$= \frac{1}{\sigma_0} \cdot \sigma_0 \left(1 - \frac{F}{F_c} \text{sgn}(v) \right) v \quad (14)$$

$$= \left(1 - \frac{F}{F_c} \text{sgn}(v) \right) v \quad (15)$$

$$(16)$$

now with $F = \sigma_0 z$ and $\text{sgn}(v)v = |v|$ we arrive at the definitive Dahl's model

$$= \left(1 - \frac{\sigma_0 z}{F_c} \text{sgn}(v) \right) v \quad (17)$$

$$\frac{dz}{dt} = v - \frac{\sigma_0 z}{F_c} |v| \quad (18)$$

[4]

2.4 Lugre Model 16

The **Lugre model** takes in consideration that coulomb friction is not constant but rather dependent of velocity, thereby replacing it with $g(v) = F_c + (F_s - F_c) e^{-|\frac{v}{v_s}|^\alpha}$ that considers the rate of change from F_s to F_c to be non-instantaneous as shown in figure 4

to create the following differential equation:

 17

3 Experiment

3.1 Drag clicking specific methods

There exists several known methods in the community to increase the accessibility of drag clicking on a mouse. With any youtube search you will find it populated with applying

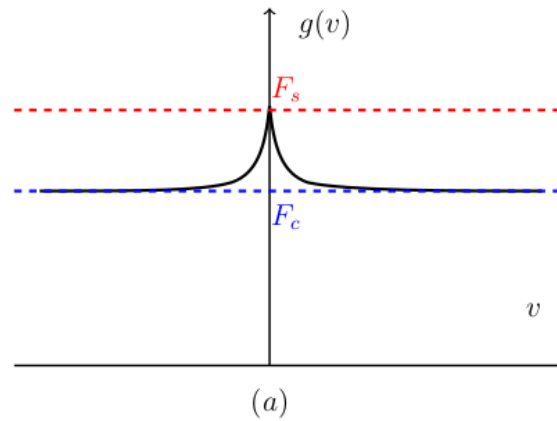


Figure 4: Graph of $g(v)$ with relative F_s and F_c coordinates

18

"tape" to your mouse, along with sharpie markers applied to that, then the mention of keeping your hands dry. In private I have discovered several new methods to increase friction, along with additive methods, I have discovered subtractive methods. Additive methods include rubbing crushed cello rosin dust onto the surface of the mouse with your finger, it was meant to adjunct slip stick properties of stringed instruments, though the effect is universal. Subtractive include rubbing the surface of your mouse with the bottom of your mousepad firmly, or with the leaf of a well hydrated plant. We will measure the rate at which static friction equalizes to coulomb friction with respect to time

19

3.2 Research Hypothesis and Variables

The difference between Coulomb friction and static friction changes depending on material surface. This investigation aims to determine how material surface affects the difference between coulomb and static coefficient of friction. It is hypothesized that Cello rosin will have the largest change due to its sticky nature, and clear PC plastic is expected with the least coefficient of friction as the surface is factory new surface polished. The independent variables are clear pc, cello rosin, and sharpie while the independent variable is the difference between static and coulomb coefficient of friction.

20

3.3 Method

The static and kinetic friction will be found by measuring steady friction force at various velocities [4]. The following diagram shows the setup:

A finger sits on a scale while taped to a string horizontally, which passes through a pulley upwards connecting to force gauge with a force exerted by a motor (Vex V5 Smart Motor 11W). The reason it is upwards is because if it were downwards, the weight of the force gauge would inherently exert a force due to gravity despite no applied force from the motor pulling it down and still displaying zero.

The experimental procedure upon procuring this setup is:

1. Clean the surface with alcohol, or reapply rosin, sharpie will not be reapplied upon initial application
2. Clean the finger with alcohol, or reapply rosin, again sharpie is neglected

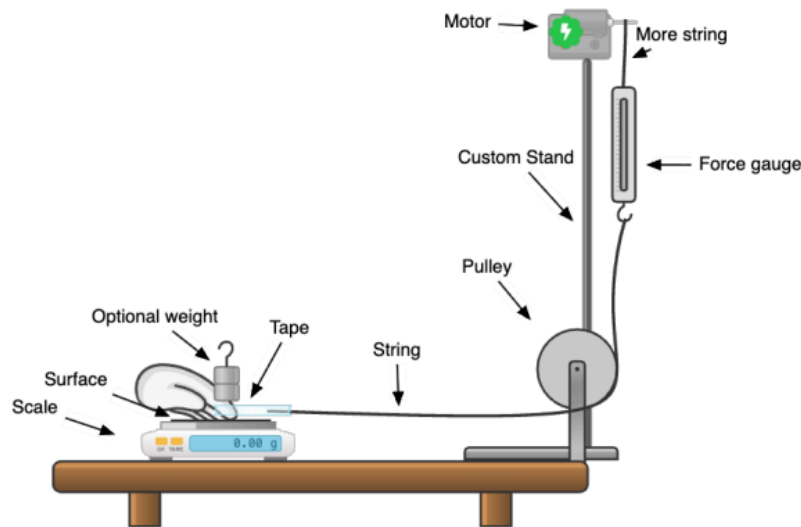


Figure 5: The experimental setup for measuring Coulomb and static friction

3. Tare the scale and while at null force, tare the force gauge
4. Insert the string into the pulley
5. Place finger onto the scale
6. While recording slow motion, ensure visibility:
 - (a) Finger state (slip/stick)
 - (b) Force gauge value
 - (c) Scale value
7. Start the motor
8. repeat steps 1-8 3 trials
9. repeat for different materials

4 Results and Discussion

4.1 Observations

The electrical tape was unable to enter kinetic friction, this is likely due to the fact that that the α in the lugre model is too large, meaning the transition to coulomb friction was too rapid (i.e. any increase in velocity resulted in a rapid decrease of frictional force to keep from sliding). It could be possible for the finger to indeed be in motion, except that the velocity is close to zero, explaining the seeming impossibility of reaching visible coulomb friction. Since the tape is viscoelastic, it has a strong elastic spring property, that like described in figure 3, the resultant frictional force is directly proportional to displacement, of which is a result of applied force, which explains the difficulty in achieving kinetic friction, and once the applied force reaches the breaking point, it slips. the addition of the elasticity of the finger introduces additional difficulty in reaching kinetic coulomb

friction since elastic on elastic contact gives extra opportunity for the reaction force to reflect the applied force.

For finger systems, it could be suggested that a modified lugre model should be used in which the σ_0 in $z = \frac{F}{\sigma_0}$ reflects the spring constant of a finger rather than microscopic bristles.

The trials of rosin exhibited slip stick of short slip distance before immediately sticking and slipping again. This could be a result of the material being non-viscoelastic, as such, the friction did not have the double spring effect, otherwise it would have doubled the spring of finger and material's displacement before finally being able to slip. Instead, it immediately jumped to coulomb in the case of sufficient applied force. This would be the result of a high α in lugre model's variation of coulomb friction as a result of velocity described in $g(v)$. And this could be a result of particulate friction that has not been looked into in this research, but the stickiness of rosin dust on the finger could be responsible for replacing the skin sebum from contacting the surface of the rosin on the pc board, and instead change into rosin on rosin contact.

4.2 Limitations of the data

The force sensor used was for measuring masses and would stop updating when no change is detected. This significantly impacted the accuracy of the data. Further improvement in methodology could involve high frequency force meter and scale that directly output into computer for high frequency data collection, in particular a computer with high monitor refreshrate and subsequently a better slow motion camera can capture much more details entailing the phases of stick slip behavior. Another limitation is the subject's muscle contraction when resetting, it is possible that force is applied prior to place the finger at the end for the string to not fall out of the pulley. It is also possible for the subject to unintentionally produce opposite force in the same direction of the frictional force to maximize slip stick behavior. Perhaps a future experiment with a moving surface rather than a moving finger would be used instead similar to [1]

References

- [1] Dennis Babu et al. “Introducing Whole Finger Effects in Surface Haptics: An Extended Stick- Slip Model Incorporating Finger Stiffness”. In: *IEEE Transactions on Haptics* 11.3 (2018), pp. 417–430. DOI: 10.1109/TOH.2018.2806458.
- [2] C. Canudas de Wit et al. “A new model for control of systems with friction”. In: *IEEE Transactions on Automatic Control* 40.3 (1995), pp. 419–425. DOI: 10.1109/9.376053.
- [3] The Efficient Engineer. *An Introduction to Stress and Strain*. 2020. URL: <https://www.youtube.com/watch?v=aQf6Q8t1FQE>.
- [4] Karl Johanastrom and Carlos Canudas-de-Wit. “Revisiting the LuGre friction model”. In: *IEEE Control Systems Magazine* 28.6 (2008), pp. 101–114. DOI: 10.1109/MCS.2008.929425.
- [5] MATLAB. *Introduction to State-Space Equations — State Space, Part 1*. 2019. URL: <https://www.youtube.com/watch?v=hpeKrMG-WP0>.

FINAL GRADE

/100

GENERAL COMMENTS

Key Areas for Improvement

1. **Missing Data:** The essay lacks quantitative results. The student describes observations but does not present measured values of friction force, normal force, or coefficients of friction for the different materials tested. This is the most critical gap.
2. **Incomplete Theory-Experiment Connection:** The student introduces sophisticated models (Dahl, LuGre) but does not clearly show how these models inform the experimental design or data analysis.
3. **Clarity of Explanation:** The student understands the concepts but struggles to explain them clearly in writing. The derivations contain notation errors, and some explanations are too condensed.
4. **Structural Issues:** Some sections (especially 2.2 and 4.1) contain sentences that trail off or are incomplete. Student appears to have changed focus mid-writing.
5. **Consistency:** There is a mismatch between the research question ("how can slip-stick friction be maximized") and the actual investigation (measuring differences between static and Coulomb friction).
6. **Formatting:** The final page contains what appears to be accidental text. The bibliography is also quite short for this topic.

PAGE 1



Comment 1

Criterion A: Framework for the Essay..

The introduction presents interesting context (Minecraft, drag clicking) but lacks a clear statement of the physics problem you are investigating. How might you more clearly establish the gap in understanding that your research aims to address, and why this matters from a physics perspective?



Comment 2

Criterion B: Terminology.

You introduce slip-stick friction as "the limit cycle of an objects' continuous oscillation between slipping and sticking." This is a good starting point, but consider: is "limit cycle" clear snought for a reader just being introduced to the topic?, or does it describe a specific mathematical behavior? How might you define/describe this advanced terminology before using it?

PAGE 2



Comment 3

Criterion A: Research Question.

Your research question is stated as: "How can slip stick friction be maximized between finger and computer mice?" However, reading further, your investigation seems to focus on studying/measuring the difference between static and Coulomb friction for different materials, not directly "maximizing" slip-stick friction. How might you refine your research question to better match what you actually investigated?



Comment 4

Criterion B: Conceptua Understanding.

You state that the Coulomb friction model suggests "frictional force is linearly proportional to the normal force." This is correct. However, you then mention that this is "perceived as due to microscopic asperities interlocking." Can you explain more precisely why friction is independent of apparent contact area despite depending on normal force? Your explanation hints at this but could be clearer. How does normal force participate in the interaction of the asperities.

Also, is the graph correct? Should not the sign of the force and the velocity be opposite?



Comment 5

Criterion B: Coinceptual understanding.

Is this clear and descriptive enough for the reader so that it is clear that you understand what you are saying? Could you expand a little bit on the reasons why discrete points will not allow instantaneous velocity and how this is relevant for the model?



Comment 6

Criterion B: Conceptual understanding.

Here, you talk about some relevant characteristics of static friction. Do you think it is relevant to offer the reader a more specific and technical definition of static friction?



Comment 7

Criterion B: Concepts – The Stribeck Effect. You introduce the Stribeck Effect but your description is brief. You mention that "we expect sharper jumps to exhibit superior drag-clicking capabilities." Can you explain why a sharper transition from static to kinetic friction would be desirable for drag clicking? What physical mechanism makes this advantageous?

PAGE 3



Comment 8

Criterion B: Terminology. You list "Ultimate Friction agin" as a phenomenon but do not explain it in this section beyond a short sentence. I think this term is not used elsewhere in your essay. Why include it? If it is not relevant to your investigation. If it is, how can you make more clear how it is relevant for your investigation?



Comment 9

Crit B: Knowing and Understanding.

You talk about "state changes". What "states" are you referring to? Are these states of matter? States of countries; states of the system, like in a complex dynamical system. If this is the latter, how can you make this clear for the reader? How can you introduce this technical term in a relevant way and support your presentation?



Comment 10

Crit B: Similar to the previous one, what are state-space equations? I think these haven't been mentioned before? Does this require a citation?



Comment 11

Crit B. Knowing and Understanding.

This seems to be describing a Dynamical System. Right now it sounds like a text taken from a book and pasted here. How can you make sure you demonstrate understanding?

Why is this a dynamical system of this type? How does this type of model help you to understand the original problem about the slip-stick friction and its connection to the use of the mouse.

Also,

You mention that state space equations "can describe the stability of a system, which is particularly useful in stick-slip motion." This is an important connection. However, you do not return to this idea in your analysis. How might you connect the concept of stability to your experimental observations (e.g., why electrical tape could not enter kinetic friction)?



Comment 12

Criterion B: Understanding Concepts.

Your introduction of the Dahl model using the "elastic bristles" analogy is helpful.

However, the transition from the bristle visualization to Equation (1) is abrupt.

I think your work can benefit from explaining the physical meaning of each parameter in Equation (1) (F_c , σ_0 , x) more explicitly before launching into the derivation. What does σ_0 represent physically, and why is it important?

PAGE 4



Comment 13

Crit. B. Knowing and understanding.

Why? Do you think explaining this may contribute to demonstrating your understanding?



Comment 14

Crit B: Knowing and understanding.

What is a state variable?

This is the same question as before in relation to explaining what a dynamical system is, in a more specific context.

Also, I can see the role of

F_c/σ_0 in equation (5), but it is not evident in eq. (8).



Comment 15

$F = -kx$

There is a negative sign missing. How can you clearly explain the connection between your model and the school elastic force?

PAGE 5



Comment 16

Criterion C: Analysis and Line of Argument. You introduce both the Dahl and LuGre models, but you do not clearly state which model you will use for your analysis, or why. Your experimental results section mentions the LuGre model's α parameter (e.g., "the α in the LuGre model is too large" for electrical tape). How does the LuGre model connect to your experimental method? Can you explain this more explicitly?



Comment 17

You introduce the LuGre model as an extension of the Dahl model where "coulomb friction is not constant but rather dependent of velocity." Can you explain what the function $g(v)$ represents physically? In particular, what is the significance of the parameters F_s , v_s , and α ? The graph in Figure 4 helps, but a more explicit explanation would strengthen your understanding.

PAGE 6



Comment 18

Criterion A: Methodology. You describe several methods for modifying the mouse surface (tape, sharpie, cello rosin, rubbing with mousepad, plant leaf). However, you do not provide a rationale for why each method might affect friction differently. For example, cello rosin is known to increase friction on string instrument bows, why might the same effect occur on a mouse? What physical mechanism are you testing?

Also, you mention "YouTube"; can you present that in a more formal way, including proper citation?

Also, "I have discovered several ways" is very informal and not a reliable way to reference your knowledge or experience.



Comment 19

Criterion C: Analysis.

You write that you will "measure the rate at which static friction equalizes to Coulomb friction with respect to time." This is a clear statement of intent. However, your experimental method section (3.3) describes measuring friction force at steady velocities, not measuring a rate of equalization over time. Is there a disconnect between what you intend to measure and what your method actually measures?



Comment 20

Criterion A: Research Methods

Your independent and dependent variables are stated. However, there is a typo: "The independent variables are clear pc, cello rosin, and Sharpie, while the dependent variable is the difference between static and coulomb coefficient of friction." I believe you meant "dependent variable" for the second case. Can you correct this and ensure your variable definitions are precise?

PAGE 7



Comment 21

Criterion C: Analysis and Findings. Your "Results and Discussion" section contains very interesting observations but no there is no quantitative data.

You describe what happened (electrical tape did not enter kinetic friction) but do not present the measured values of friction force, normal force, or coefficient of friction for the different materials. Where are your data? Without presenting and analyzing actual measurements, it is difficult to draw meaningful conclusions about which material maximizes slip-stick friction.



Comment 22

Criterion D: Discussion and Evaluation This observation about "electrical tape was unable to enter kinetic friction" is interesting and central to your investigation. You suggest this is because "the α in the LuGre model is too large, meaning the transition to coulomb friction was too rapid." This is a sophisticated explanation. However, you do not present any clear analysis of the effect of α in the model or measure measured values of α . How do you know it is "too large" without calculating or estimating it? What evidence supports this claim?

Again, this seem to be important, you could dig deeper in this aspect.



Comment 23

Criterion D: Discussion and Evaluation.

You mention that "the tape is viscoelastic" and "has a strong elastic spring property." This connects to your earlier Dahl model discussion (analogy in Figure 3 and maybe Figure 2).

Can you develop this connection more explicitly?

How does the viscoelasticity of the tape relate to the Dahl model's parameters (σ_0 , F_c)?

This would strengthen your analysis significantly.

PAGE 8

PAGE 9